Pith

The central portion is occupied by a large pith. It consists of thin-walled parenchyma cells with intercellular spaces. These cells are filled with abundant starch grains.

Primary Structure of Dicot Stem (Sunflower stem)

The transverse section of the dicot stem (Sunflower) shows the following plan of arrangement of tissues from the periphery to the centre.

Epidermis

It is protective in function and forms the outermost layer of the stem. It is a single layer of parenchymatous rectangular cells. The cells are compactly arranged without intercellular spaces. The outer walls of epidermal cells have a layer called cuticle. The cuticle checks the transpiration. The cuticle is made up of waxy substance known as cutin. Stomata may be present here and there. A large number of multicellular hairs occur on the epidermis.

Cortex

Cortex lies below the epidermis. The cortex is differentiated into three zones. Below the epidermis, there are few layers of collenchyma cells. This zone is called **hypodermis.** It gives mechanical strength of the Stem. These cells are living and thickened at the corners.

Inner to the hypodermis, a few layers of chlorenchyma cells are present with conspicuous intercellular spaces. This region performs photosynthesis. Some resin ducts also occur here. The third zone is made up of parenchyma cells. These cells store food materials. The innermost layer of the cortex is called **endodermis**. The cells of this layer are barrel shaped and arrange compactly without intercellular spaces. Since starch grains are abundant in these cells, this layer is also known a **starch sheath**. This layer is morphologically homologous to the endodermis found in the root. In most of the dicot stems, endodermis with casparian strips is not developed.

Stele

The central part of the stem inner to the endodermis is known as **stele**. It consists of pericyle, vascular bundles and pith. In dicot stem, vascular bundles are arranged in a ring around the pith. This type of stele is called **eustele.**

Pericycle

Pericycle is the layers of cells that occur between the endodermis and vascular bundles. In the stem of **sunflower (Helianthus),**a few layers of sclerenchyma cell occur in patches outside the phloem in each vascular bundle. This patch of sclerenchyma cell is called **Bundle cap or Hardbast**. The bundle caps and the parenchyma cells between them constitute the pericycle in the stem of sunflower.

Vascular Bundles

The vascular bundles consist of xylem, phloem and cambium. Xylem and phloem in the stem occur together and form the vascular bundles. These vascular bundles are **Wedge shaped**. They are arranged in the form of a ring. Each vascular bundle is **conjoint, collateral, open and endarch**.

Phloem

Phloem consists of sieve tubes, companion cells and phloem parenchyma. Phloem fibres are absent in the primary phloem. Phloem conducts organic food materials from the leaves to other parts of the plant body.

Cambium

Cambium consists of **brick shaped** and thin walled meristematic cells. It is one to four layers in thickness. These cells are capable of forming new cells during **secondary growth**.

Xylem

Xylem consists of xylem fibres, xylem parrenchyma vessels and tracheids. Vessels are thick walled and arranged in a few rows. Xylem conducts water and minerals from the root to the other parts of the plant body.

Pith or medulla

The large central portion of the stem is called **pith**. It is composed of parenchyma cells

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